GP Meeting: 13 July 2026

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July 2026

1 GP-Internal Feature Analysis

1.1 Motivation

The existing features describe the input data in terms of morphology, roughness, signal-to-noise ratio (SNR), and sampling characteristics. They therefore characterize the type and quality of the data provided to the GP.

In contrast, GP-internal features describe the relationship between the GP's predictive uncertainty and its actual prediction error. This relationship is important for distinguishing the following four cases:

- **Case 1: Large mean prediction error.** If the predictive uncertainty is also large, the GP may correctly recognize that the region is difficult to predict. If the uncertainty remains small, the GP has both an inaccurate mean prediction and a calibration problem. Post-hoc uncertainty scaling cannot correct the inaccurate predictive mean.
- **Case 2: Small mean prediction error but excessively large uncertainty.** This indicates an underconfident prediction: the GP predicts the mean reasonably well but produces an unnecessarily wide predictive distribution. Post-hoc uncertainty shrinkage may improve calibration in this case.
- **Case 3: Small mean prediction error but insufficient uncertainty.** This indicates an overconfident prediction. Post-hoc uncertainty inflation may improve calibration.
- **Case 4: Prediction error and predictive uncertainty agree.** In this case, the object is approximately well calibrated. At the aggregate level considered here, this corresponds to a standardized error magnitude near one.

1.2 Experiment

To examine these cases, each object-band pair is represented as one point in a scatter plot. The horizontal axis shows the root-mean-square predictive un-

certainty over the held-out observations, while the vertical axis shows the root-mean-square error (RMSE) of the predictive mean:

$$\text{Predictive uncertainty scale} = \sqrt{\text{mean}\left(\sigma^2_{\text{pred}}\right)}$$

$$\text{RMSE} = \sqrt{\text{mean}\left(y_{\text{true}} - \mu_{\text{pred}}\right)^2}$$

A diagonal reference line, $y = x$, is added to the plot. Points close to this line have prediction errors whose overall magnitude is comparable to the predicted uncertainty. Points above the line have errors larger than their predicted uncertainties and therefore indicate an overconfident tendency. Points below the line have uncertainties larger than their errors and indicate an underconfident tendency.

The results are shown in Figures 1a and 1b.

Most points lie below the diagonal and are associated with relatively small standardized errors. This indicates that many of the fitted GPs are underconfident: their predictive uncertainties are larger than the observed prediction errors.

The extreme red point in Figure 1b represents a pathological overconfident case. The corresponding light curve is shown in Figure 2. The GP fails to reconstruct a sharp peak that is supported by only one high-flux observation. Although this observation is included in the training set, a single point is insufficient to constrain the width and shape of the peak. The GP consequently produces a smooth prediction that misses the held-out measurements near the peak while assigning insufficient uncertainty to this failure.

2 MOGP Cross-Band Study

The main research question is:

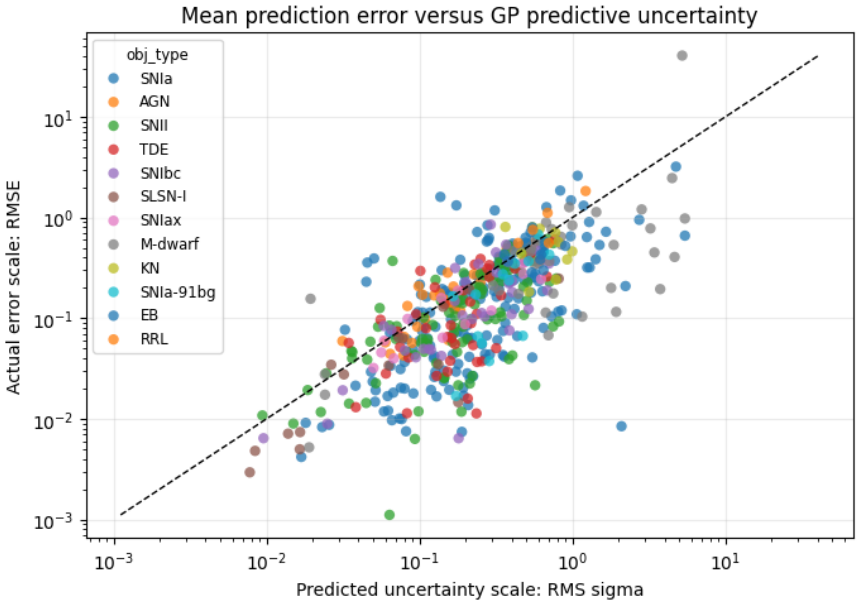
> Can cross-band predictive gain be explained more effectively by class-specific band similarity than by central-wavelength separation alone?

2.1 Class-Level Inter-Band Distance Analysis

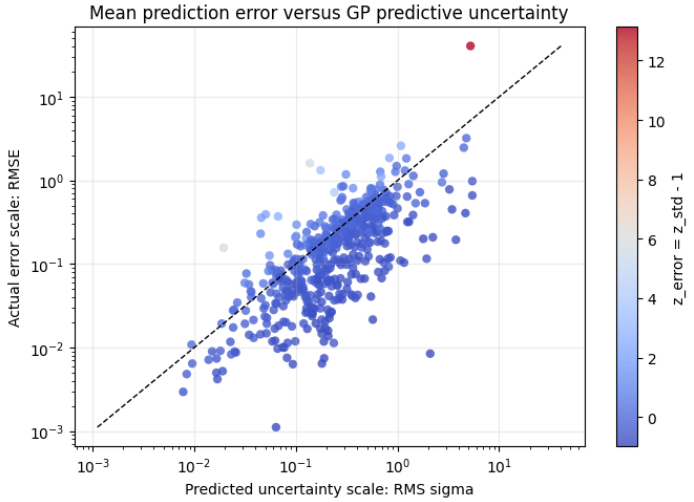
The wavelength kernel assumes that, all else being equal, bands with similar central wavelengths should have more strongly correlated latent function values:

$$|\lambda_b - \lambda_{b'}| \downarrow \quad \Rightarrow \quad k_\lambda(\lambda_b, \lambda_{b'}) \uparrow$$

A large kernel covariance implies that the latent light-curve values at the two wavelengths are expected to be similar. Because this assumption concerns the similarity of the function values themselves, an amplitude-sensitive functional distance is more directly related to the wavelength kernel than a correlation



(a) Object-band points labelled by transient class.



(b) Object-band points coloured by the standardized error metric z_{error}.

Figure 1: Comparison between the predictive uncertainty scale and the RMSE of the predictive mean.

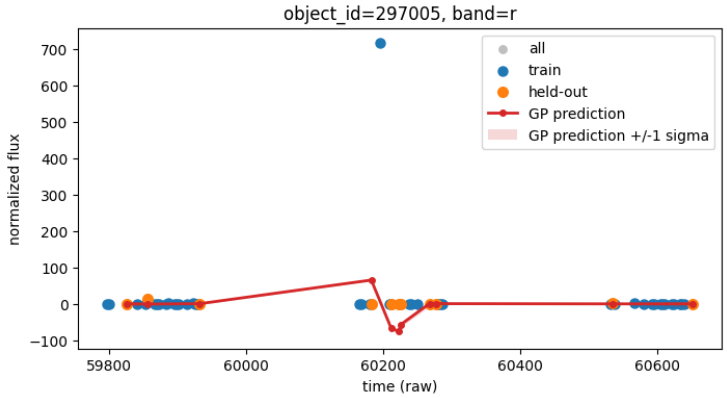


Figure 2: Predicted light curve for the object corresponding to the extreme red point in Figure 1b.

coefficient. Correlation measures similarity in shape but is insensitive to absolute amplitude differences, whereas the MOGP must model both the shapes and relative amplitudes of the light curves.

I therefore use an object-level inter-band distance to measure how closely two bands follow one another within the same object. For object n and bands b and b', the distance is defined as

$$D_{n,bb'} = \sqrt{\text{mean}\left(f_{n,b} - f_{n,b'}\right)^2}$$

where $f_{n,b}$ and $f_{n,b'}$ are the aligned, normalized, and binned light curves. The distance is computed using only time bins for which both bands have valid aggregated flux values.

The object-level distances are then aggregated within each class using the median:

$$D_{c,bb'} = \text{median}_{n \in c} D_{n,bb'}$$

Here, c denotes the transient class. This quantity measures whether two bands tend to have similar light curves within individual objects belonging to the same class.

Because the observations are irregular and asynchronous, the following preprocessing steps are applied before computing the distances:

1. For each object, observation times are shifted relative to the object's estimated peak time.

2. All bands of the same object are normalized using a shared robust flux scale. The scale is defined as the median of the five largest absolute flux values across all bands of that object.

3. The relative-time range is divided into common bins using `np.linspace(-500, 500, 50)`. The bin centres are defined as the averages of the corresponding left and right bin boundaries.

4. Within each object and band, all observations falling into the same time bin are aggregated using their median flux.

5. For every band pair, the root-mean-square difference is computed over the bins in which both bands are available.

6. The resulting object-level distances are aggregated across objects within each class using the median.

This procedure preserves class-specific differences in both light-curve shape and relative band amplitude while reducing the influence of irregular sampling, extreme flux values, and individual pathological objects.

For visualization, I created a heatmap of the inter-band distance matrix for each class and a scatter plot comparing wavelength separation with inter-band distance. The results are shown in Figures 3 and 4.

Several supernova classes exhibit the expected wavelength-ordered pattern. For SNIa, SNII, SNIbc, SLSN-I, SNIax, and SNIa-91bg, distances are generally smaller near the diagonal of the heatmaps and increase as the wavelength separation becomes larger. TDE also shows a reasonably smooth relationship. These results are broadly consistent with the assumption underlying the wavelength kernel:

$$|\lambda_b - \lambda_{b'}| \downarrow \quad \Rightarrow \quad D_{c,bb'} \downarrow$$

In contrast, AGN, M-dwarf, KN, EB, and RRL show weak or non-monotonic relationships between wavelength separation and empirical inter-band distance. In some of these classes, neighbouring bands, such as u and g, are among the most dissimilar band pairs. For these classes, a stationary wavelength kernel based only on $|\Delta\lambda|$ may therefore provide an incomplete description of the empirical cross-band relationships.

These observations motivate the following hypothesis:

> Classes for which empirical inter-band distance increases more consistently with wavelength separation should benefit more from wavelength-based cross-band information transfer.

To test this hypothesis, I will compare the no-transfer model with the MOGP separately for each class. The predictive gain will be evaluated using ΔRMSE, ΔNLPD, and ΔCRPS.

The change in NLPD will additionally be decomposed into contributions associated with the predictive mean error and the predicted variance. This will help distinguish improvements caused by more accurate mean predictions from improvements caused by better uncertainty estimation.

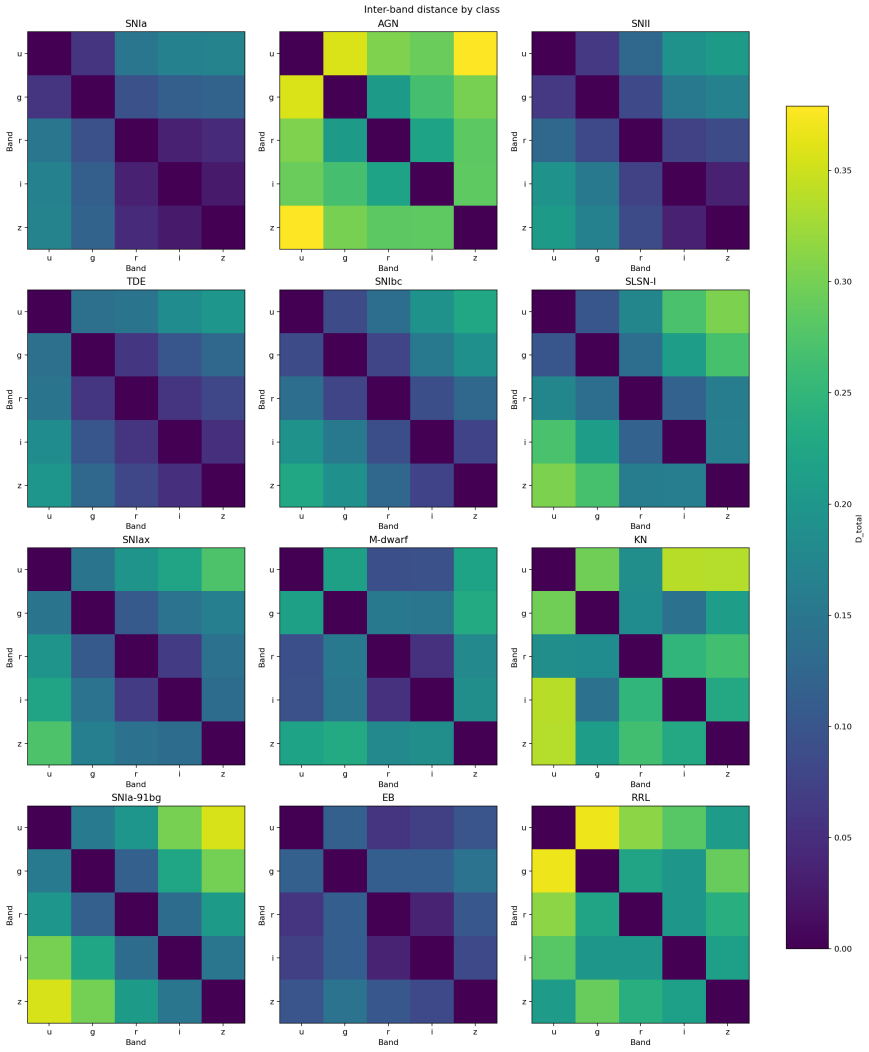


Figure 3: Class-specific inter-band distance matrices. Smaller values indicate greater empirical similarity between the normalized light curves of two bands.

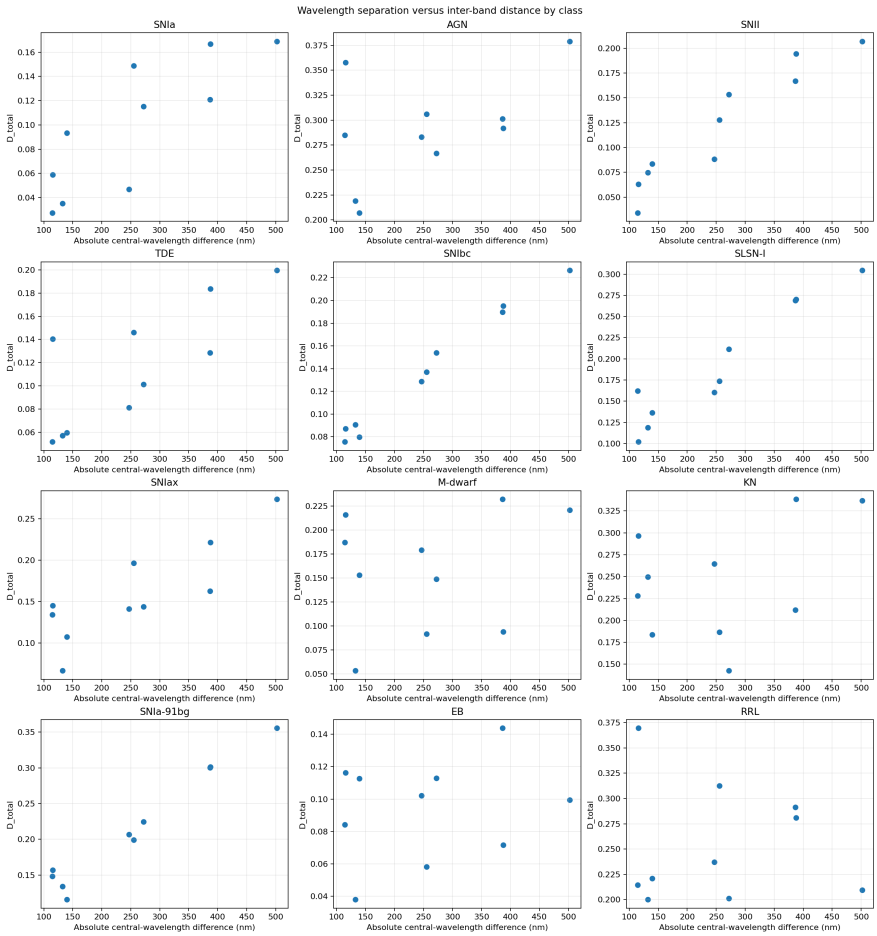


Figure 4: Relationship between central-wavelength separation and empirical inter-band distance for each transient class.

2.2 Planned Cross-Band Gain Analysis

For each class and held-out target band, I will measure the difference between the no-transfer baseline and the MOGP. These predictive gains will then be compared with the corresponding empirical inter-band distances and wavelength separations.

The purpose of this analysis is not only to determine whether the MOGP improves prediction, but also to examine whether its improvement is largest for classes whose empirical band relationships agree with the assumptions encoded by the wavelength kernel.

Specifically, for each class, compare the wavelength-distance compatibility and cross-band predictive gain. The formula is listed below:

$$C_c = \rho_{\text{spearman}}(|\Delta\lambda|, D_{\text{total}})$$

versus

$$G_c = \text{median}\,(\text{NLPD}_{\text{no transfer}} - \text{NLPD}_{\text{MOGP}})$$

Note: This get deleted!!!

We note from section 2.1 that band u and g are the neighboring bands with the inter-band distance being small for supernova classes while being large for some other classes. Therefore, we use flux from target band u with g band as the auxiliary band to test and compare the performance of MOGP and independent-band-control MOGP model. The class-level visualization is shown in Figure 5.

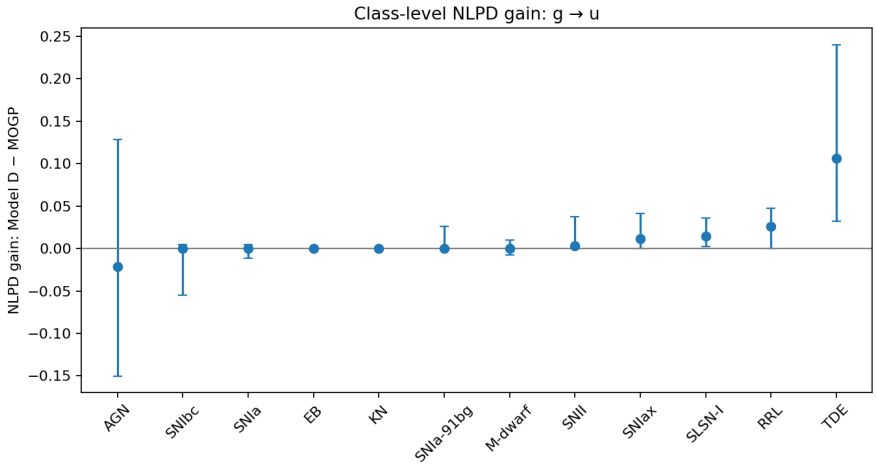
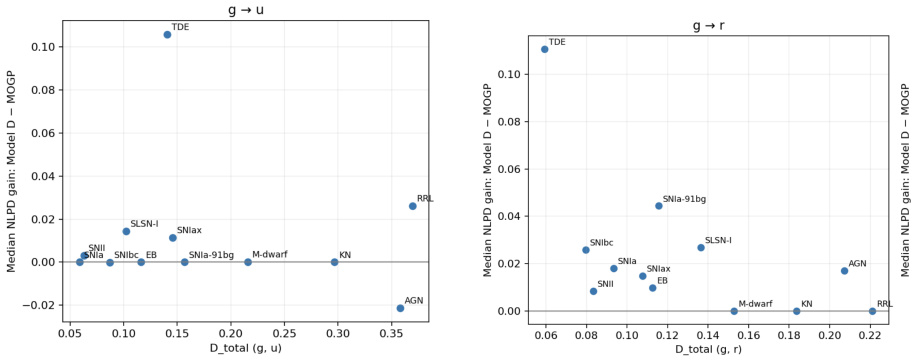


Figure 5: Class-level NLPD gain.

The pair-specific $g \to u$ analysis shows no clear monotonic relationship between the class-level inter-band distance, $D_{c,gu}$, and the predictive gain from cross-band transfer, visualized in Figure 5, $\Delta\text{NLPD} = \text{NLPD}_D - \text{NLPD}_{\text{MOGP}}$.

To further explain the experiment results, consider decomposing the NLPD. For a Gaussian predictive distribution $p(y_i \mid x_i) = \mathcal{N}(\mu_i, \sigma_i^2)$, the negative log predictive density for observation i is



(a) Class-level NLPD gain vs. distance (target band u, auxiliary band g).

(b) Class-level NLPD gain vs. distance (target band r, auxiliary band g).

Figure 6: Overall caption describing both subfigures combined.

$$\mathrm{NLPD}_i = -\log p(y_i \mid x_i) = \frac{1}{2}\log(2\pi\sigma_i^2) + \frac{1}{2}\frac{(y_i-\mu_i)^2}{\sigma_i^2}.$$

This expression contains two contributions. The first term,

$$\frac{1}{2}\log(2\pi\sigma_i^2),$$

is the sharpness term. Sharpness describes how concentrated the predictive distribution is: a smaller predictive variance σ_i^2 corresponds to a narrower and therefore sharper distribution, whereas a larger variance corresponds to a broader and less sharp distribution. The second term,

$$\frac{1}{2}\frac{(y_i-\mu_i)^2}{\sigma_i^2} = \frac{1}{2}z_i^2, \qquad z_i = \frac{y_i-\mu_i}{\sigma_i},$$

is the standardized squared-error term. It measures the discrepancy between the predictive mean and the observation relative to the predicted uncertainty. Thus, NLPD balances the concentration of the predictive distribution against whether the observed error is consistent with the stated uncertainty. A sharp prediction reduces the first term, but it can strongly increase the second term if the predictive mean is inaccurate.

The NLPD gain can be decomposed as

$$G_{\mathrm{NLPD},i} = \underbrace{\frac{1}{2}\log\left(\frac{\sigma_{B,i}^2}{\sigma_{T,i}^2}\right)}_{G_{\mathrm{sharp},i}} + \underbrace{\frac{1}{2}\left[\frac{(y_i-\mu_{B,i})^2}{\sigma_{B,i}^2} - \frac{(y_i-\mu_{T,i})^2}{\sigma_{T,i}^2}\right]}_{G_{\mathrm{std.err},i}}.$$

Here, $G_{\mathrm{sharp},i}$ measures the change in predictive concentration, while $G_{\mathrm{std.err},i}$ measures the change in standardized prediction error. The decomposition is

exact, but the two terms are not causally independent because changing the predictive variance also changes the standardized-error term.

From Figure 14, for TDEs, the positive median gain (0.106) is driven primarily by improved sharpness, while a small standardized-error penalty indicates that the MOGP produces narrower predictive distributions without substantially improving the predictive mean. AGN exhibits the opposite behavior: although the MOGP is sharper, its larger standardized-error penalty and slightly worse RMSE produce a negative median NLPD gain (-0.021), consistent with overconfidence or negative transfer. RRL shows a modest positive median gain (0.026), potentially supported by dense temporal coverage, despite having a large $D_{c,gu}$; however, this conclusion is tentative because only five objects are available and one performs substantially worse. Overall, inter-band template distance alone does not determine transfer effectiveness: temporal support, uncertainty calibration, class-specific variability, and object-level heterogeneity also play important roles.

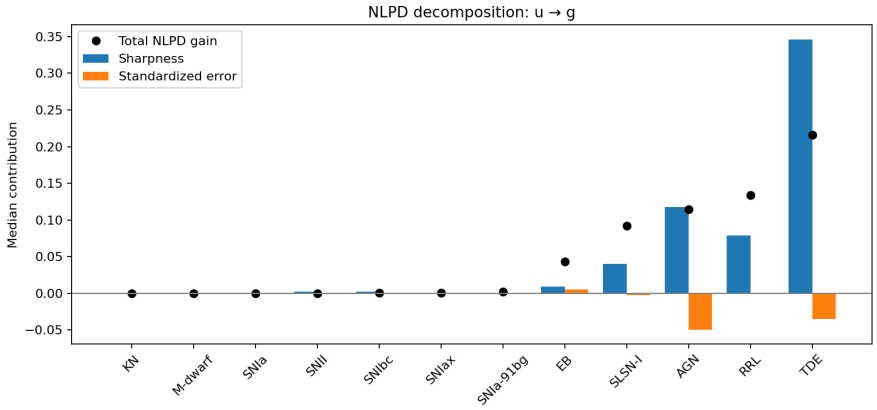


Figure 7: Class-level NLPD gain decomposition (band u to g).

Note! change it to class-level mean instead of median!!!

2.3 Specific Comparison of MOGP and Independent-band-control MOGP

The comparison is done on objects from class M-dwarf, with r as the target band and the ratio of g and i being 0.5 and 1.0. The improvement of NLPD is computed and sorted in an ascending order. The worst performance improvement from two objects are selected and investigated. Below figures show two failure cases.

The examined failure cases show that cross-band transfer can introduce negative transfer when prominent auxiliary-band features are not consistently expressed in the target band. The NLPD degradation is almost entirely attributable to the standardized-error component, while the sharpness contribu-

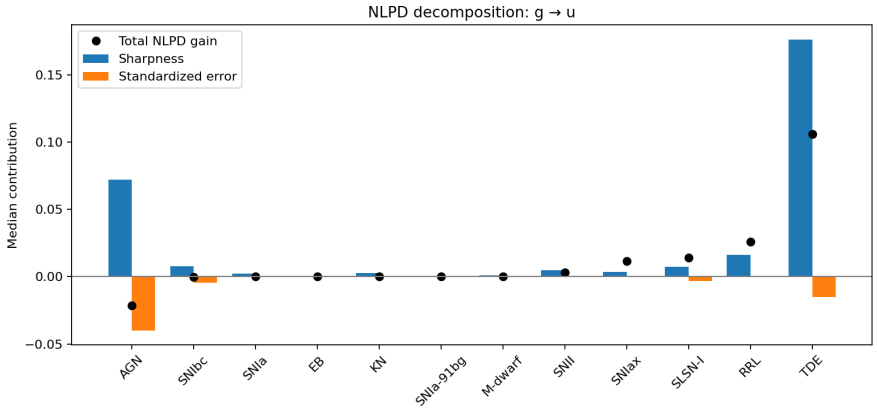


Figure 8: Class-level NLPD gain decomposition (band g to u).

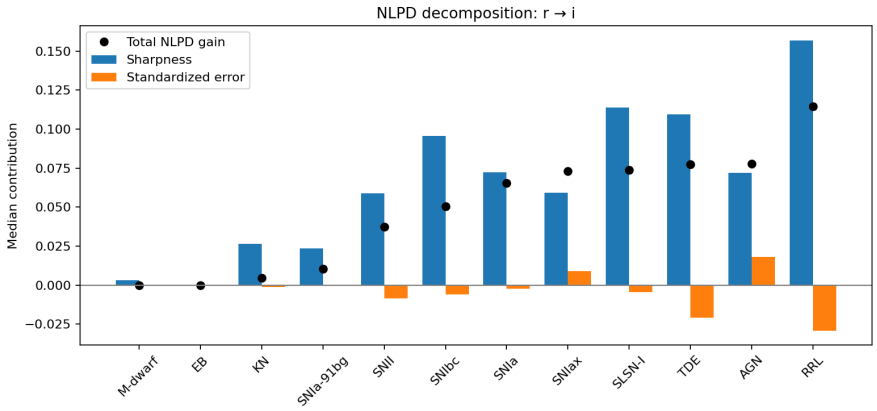


Figure 9: Class-level NLPD gain decomposition (band r to i).

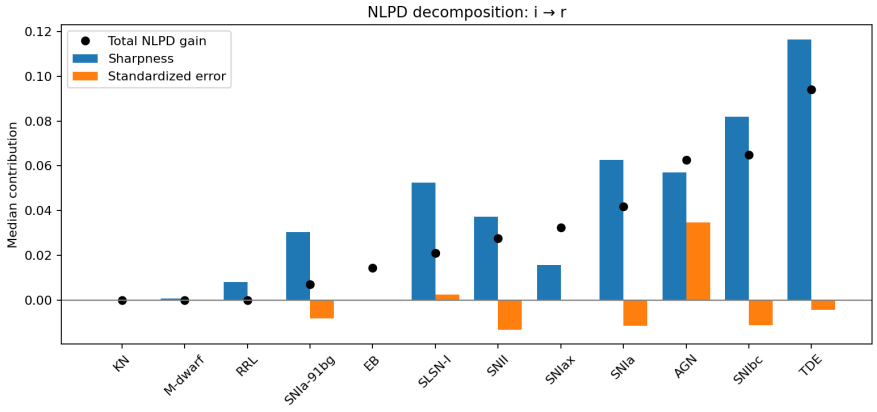


Figure 10: Class-level NLPD gain decomposition (band i to r).

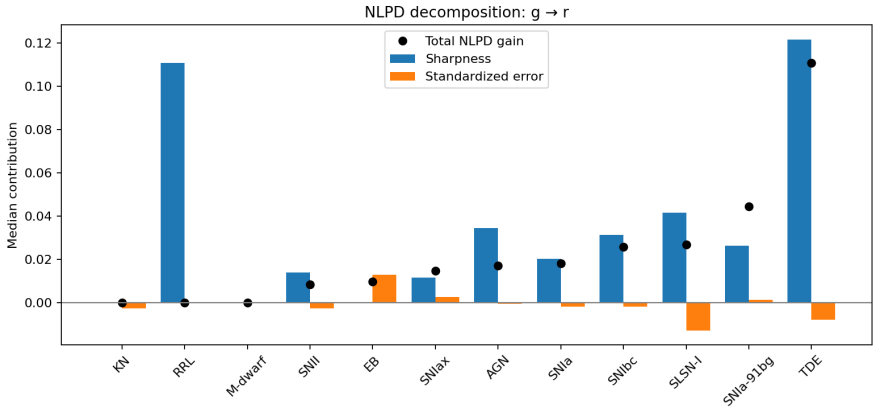


Figure 11: Class-level NLPD gain decomposition (band g to r).

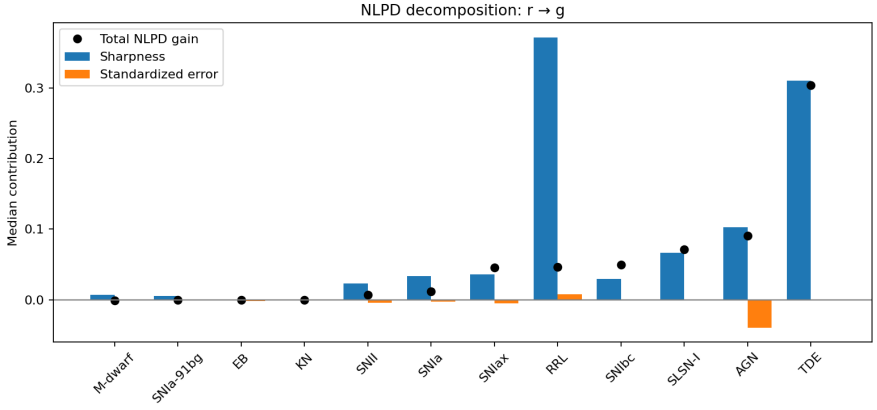


Figure 12: Class-level NLPD gain decomposition (band i to z).

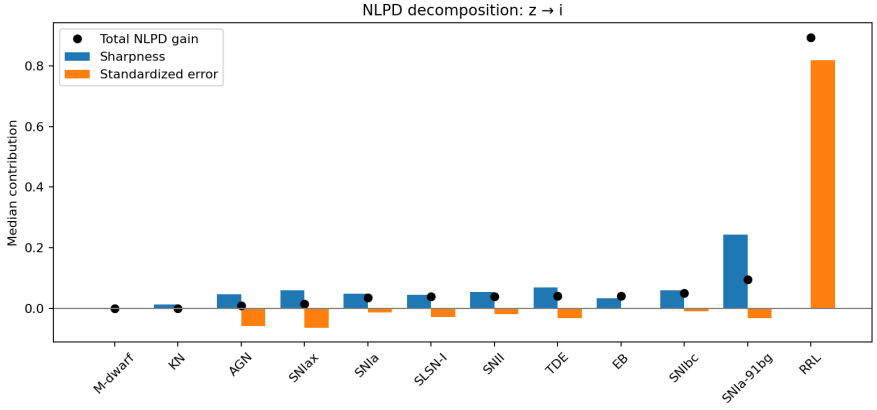


Figure 13: Class-level NLPD gain decomposition (band z to i).

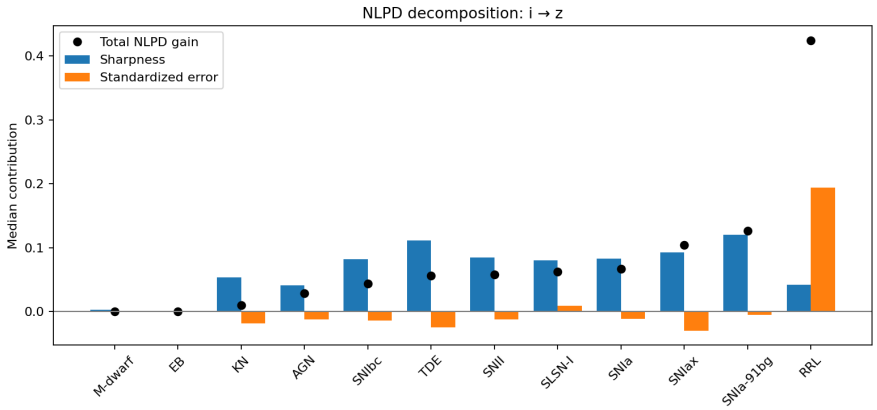
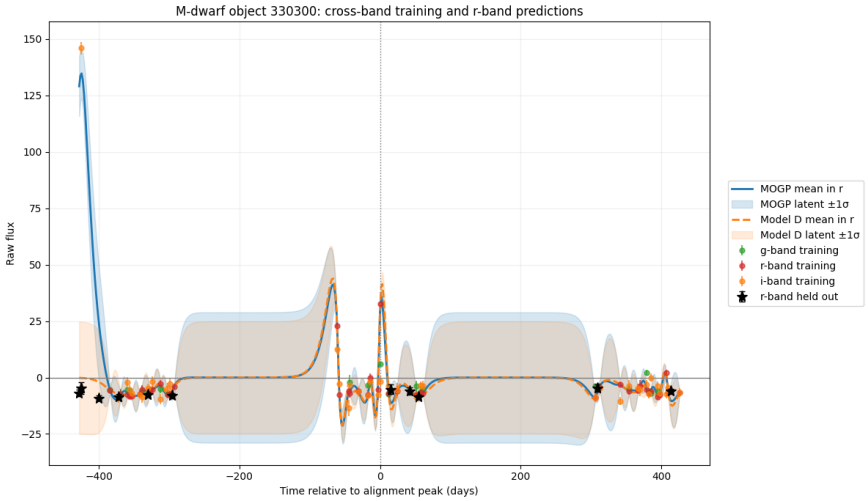
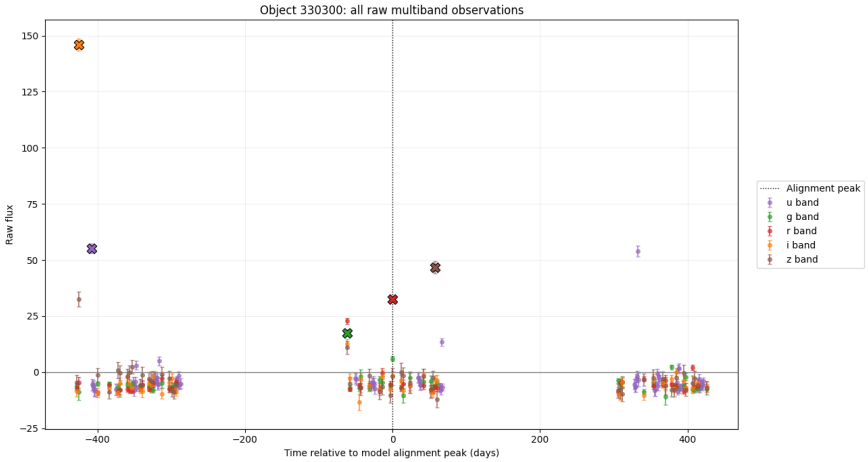


Figure 14: Class-level NLPD gain decomposition (band i to z).

tion is small, indicating that the principal failure is distortion of the predictive mean rather than a change in predictive concentration. For extremely rapid and sparsely sampled events, both MOGP and the band-independent model can fail because the assumed smooth temporal kernel cannot adequately represent the observed variability. Consequently, the benefit of MOGP depends not only on the availability of auxiliary observations, but also on the object-specific consistency of the selected band pair and the suitability of the temporal kernel.

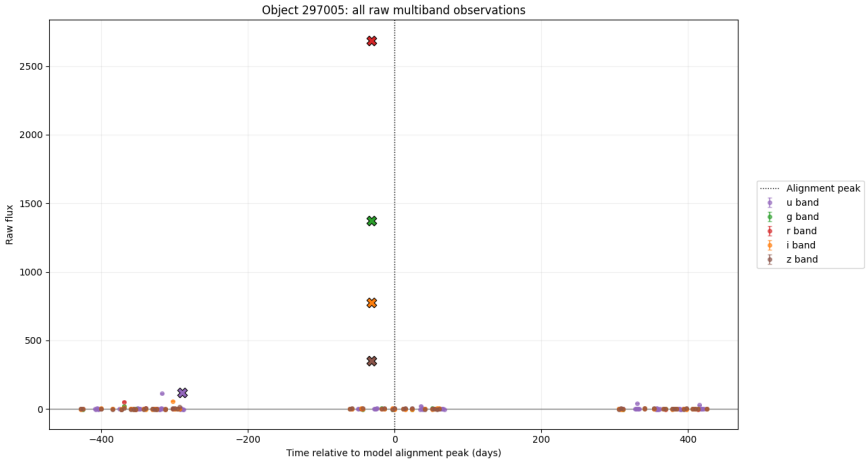


(a) Cross-band training and r-band predictions for object 330300.

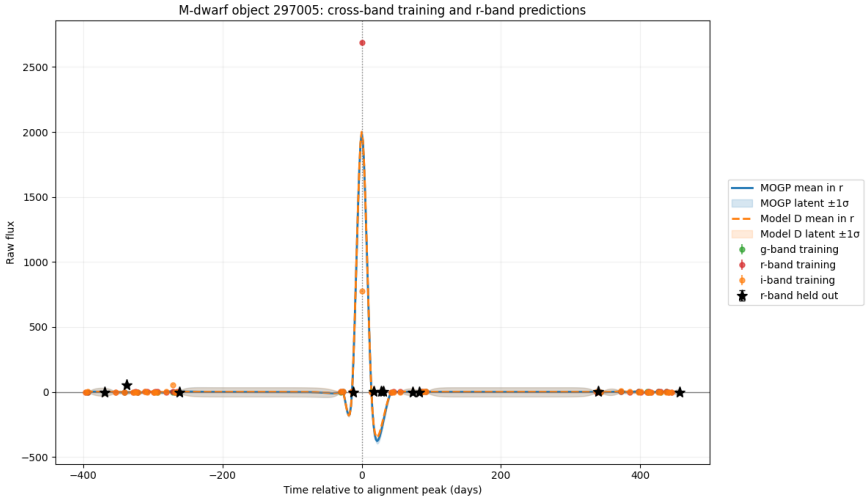


(b) Raw multiband observations for object 330300.

Figure 15: The first-worst NLPD improvement among the M-dwarf objects, corresponding to object 330300.



(a) Raw multiband observations for object 297005.



(b) Cross-band training and r-band predictions for object 297005.

Figure 16: The third-worst NLPD improvement among the M-dwarf objects, corresponding to object 297005.